



Computer networking summary notes

Computer Networking (Western Sydney University)



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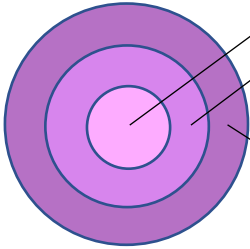
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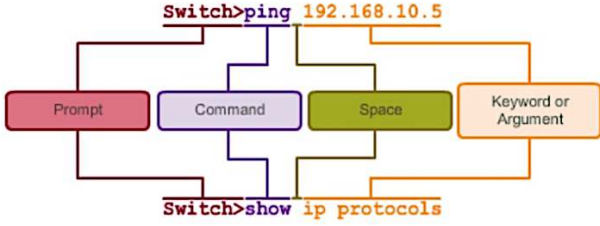
Computer Networking Questions/Answers

LECTURE 1: Introduction to Networking

How does technology today compare with the past?	- Immediate nature of communication through social network platforms to encourage global communities - Allows access to vast quantities of information globally - Supports the way we work & play (e.g. Online classes)	
Describe the different sizes of networks.	- Small home – Few devices, connects to internet - Small office – Device connects to corporate network - Medium to Large – Many locations, 100s-1000s devices - Worldwide – Connects millions of devices (e.g. Internet)	
Identify what is a client and server.	Client: Computers (end device) which sends requests to a server for the retrieval of information Server: Computer which provide the information requested by an end device on a network	
Describe and evaluate Peer-to-Peer networking.	- The client also functions as a server (e.g. Bit torrent) <u>Advantages</u> • Easy Setup • Uncomplex • Low cost <u>Disadvantages</u> • Administration not central • Insecure • Slow performance • Not scalable	
Identify and explain the 3 categories of components of network infrastructure.	1. <u>Devices</u> - End devices: Send and receive messages - Intermediary Network Devices: Interconnects end devices to form internetwork, retransmit signals, maintain pathways info, notify of failures + error 2. <u>Media</u> - Metallic Cable (copper): Electrical pulses - Optic fibre (glass): Light waves - Wireless: Wavelengths from EM Spectrum 3. <u>Services</u> - Network Applications (e.g. email/web hosting services) - Processes: Provides functionality to move/direct message	
Outline how a network is represented.	• Visually shown by Topology diagrams (2 types) - Physical: Shows the device's physical location and cabling - Logical: Identifies devices, ports, and address schemes	
Outline and compare LAN and WAN	<u>LAN (Local Area Network)</u> - Small area (e.g. School) - Managed by individual - High bandwidth speed	<u>WAN (Wide Area Network)</u> - Interconnected LAN's - Multiple service provider operated - Slower speeds
	Other network types: - LAN < MAN < WAN - Storage AN: File server, data storage, retrieval - Wireless LAN	
Compare and contrast the Internet, Intranet, and Extranet	<u>Internet</u> - Worldwide - Interconnected WANs - Public (not owned) - Infrastructure maintained by: IETF, ICAAN, IAB, etc.	<u>Intranet</u> - Similar to <i>Internet</i> - Private to organisations <u>Extranet</u> - Provide secure access to the network

LECTURE 1: Introduction to Networking	What technologies today do we use to access the internet?	- DSL: High bandwidth, Runs over phone line, (down > up) - Cable: High bandwidth, From TV provider - Cellular: Cell phone network, Depends on signal - Satellite: Clear line of sight, Rural areas - Dial-up Telephone: Cheap, Low bandwidth	
	How do businesses and organisation access the internet differently to home users?	• Business's may need higher bandwidth & faster connections to support daily activities efficiently. - Leased line: Reserved circuits, - Ethernet WAN - DSL: SDSL (Download = Upload [speeds]) - Satellite: Usually when wired not available	
	Describe the differences of a converging network to a traditional.	- Converging has multiple services on one link (data, voice, video, etc.) - 1 link ∴ same set of rules/standards	
	Explain four characteristics of a network's infrastructure contributes to the successfulness and reliability of a network?	Fault tolerance - Multiple pathways for data travel (routers) - Limits number of devices affected Scalability (Expands with ease & speed to cater new users) Quality of Service - Managing flow of data/traffic for reliable content delivery - E.g. Buffering of videos Security - <u>Network infrastructure security</u> • Physically safe • Safe in software (authorization levels) - <u>Information security</u> • Data/information is protected	
	Identify some trends which have appeared within market	- BYOD (Bring Your Own Device) - Online collaboration - Video communication (Facetime, skype) - Cloud computing	
	Outline the types of clouds and how cloud computing is being used.	• Uses: Data storage, Apps via cloud, Easy access Public: Free/Pay-per-use Private: Used by specific organisations or entity Hybrid: Uses 2/+ Cloud types Custom (E.g. Government)	
	Identify and outline two networking technologies used in homes.	Powerline Networking - Uses existing electrical wiring to connect devices Wireless Broadband - WISP: Connects subscribers to designated access points or hotspots. Antenna/Dish is installed to connect.	
	Identify types of threats to security.	External - Virus, worm, trojan - Spyware/adware - Hacking - Denial of Service - Identity/data theft	Internal - BYOD increase chances - Internal users cause most security breaches
	Identify solutions to mitigate security threats	Small Home Networks - Antispyware & Antivirus - Firewall	Organisation Larger Networks - Dedicated firewall - Access control list - Intrusion prevention systems - VPN

LECTURE 2: Configure a Network Operating System	Define an operating System and outline its three layers.	- Operating system is software that manages computer hardware and software. (All end-devices need one)  Hardware: Physical part of computer. OS Kernel: Communicates with hardware (directly). OS Shell: Command-line interface (CLI) or graphical-user interface (GUI) to interact with applications.	
	Identify examples for the purpose of an OS.	GUI-based PC OS (computers) allows: - Use of mouse to make selections - Entering of text - View output on a monitor CLI-based Network OS (switches, router) allows: - Use of keyboard to run CLI-based programs - Enter text & text-based commands <i>Note: All devices come with a default IOS and features (upgradeable)</i>	
	Discuss common methods (3) to access IOS (Internetwork Operating System).	Console port: Physical management port providing <i>out-of-band</i> (dedicated management channel) access to device. - Able to access when no networking services configured Secure Shell (SSH): <i>In-band method</i> to establish remote <i>secure</i> CLI connection through virtual interface over network. - Needs active networking services on device. - Recommended method for remote management because secure (encrypted authentication) Telnet: <i>In-band method</i> to establish remote <i>insecure</i> CLI connection through virtual interface over network. - User authentication is sent over network in plain text. - Best practice to use SSH for remote management (secure).	
	What is a terminal Emulation Program?	Application to make the connected device appear like a terminal or client computer networked to a server/mainframe. E.g. Tera Term, PuTTY, SecureCRT, OS X Terminal.	
	Describe the two Primary Command modes and how to navigate between these modes.	User EXEC Mode - Limited capabilities of basic monitoring commands - "view-only" mode - Prompt ending with '>'	Privileged EXEC Mode - All commands available - Execute configuration commands - Prompt ending with '#'
		Use command "enable" to access <i>Privileged EXEC mode</i> and "disable" to return to <i>User EXEC mode</i>.	
	Describe global configuration and specific sub-configuration modes.	- CLI configuration changes from Global Configuration mode affect operation of device as a whole. - Use command "configure terminal" to enter <i>global config</i> and "end" to exit <i>global config</i> from any configuration mode.	
		- Sub-configuration allows configuration of a specific part or function of the device. Common Sub-configuration modes: • Interface mode: Config. Network interface • Line mode: Config. Console, AUX, Telnet, SSH access - Sub-configuration modes can be entered after <i>Global config mode</i> is accessed. - Use "exit" to move from specific mode to previous general mode.	

LECTURE 2: Configure a Network Operating System	Outline the basic IOS Command structure.	- All IOS command have a specific format/syntax.  - Keyword: Specific parameter defined in OS (e.g. ip protocols) - Argument: Not predefined (value/variable defined by user)
	Identify and describe IOS command conventions.	- When describing the use of commands, we generally use these conventions: boldface: Boldface text indicates commands and keywords that you literally enter as shown. <i>italics:</i> Italic text indicates arguments for which you supply values. [x]: Square brackets indicate an optional element. {x}: Braces indicate a required element. [x {y z}]: Braces and vertical lines within square brackets indicate a required choice.
	Describe the two IOS help features.	IOS Command Syntax Check - Checks the entered command (from left to right) to determine action requested. - If error, IOS provides feedback (e.g. Ambiguous/ Incomplete/ Incorrect command etc.) IOS Context Sensitive Help - Provide list of command & arguments associated with those. - Use '?' to access help.

LECTURE 3: Network Protocols and Communication	Identify three common aspects that all methods of communication have in common.	- Source/Sender: People/devices that send the message - Destination/receiver: Receives message and interprets it - Channel: Media that provides pathway for message to travel from source to destination.
	Identify protocols/rules necessary for effective communication.	- Rules/Protocols govern all forms of communication: 1. An identified sender and receiver 2. Common language and grammar 3. Speed & timing of delivery 4. Confirmation/acknowledgement requirements
	Discuss protocols (5) used for messages in addition to the fundamental traits in network communications. (<i>encoding, formatting & encapsulation, size, timing, delivery options</i>)	Message encoding: Process of converting message into acceptable format (bits) for transmission between two hosts. - Each <i>bit</i> is encoded into a pattern of sound, electrical or light. - Sender's message will be encoded into <i>bits</i> before sent, receiver will receive this and decode it to interpret message.
		Message formatting & encapsulation: Each message is formatted and encapsulated in a specific format before sent. (e.g. Letter) - Message formats depend on type of message and channel. - Encapsulated in a specific format called <i>frame</i>.
		Message size: Breaking long messages into smaller pieces to travel across network. The receiving host reconstructs original message. - Each <i>piece</i> is sent in a separate <i>frame</i> containing addressing information
		Message timing: Rules of engagement - <u>Access method</u> (when to send messages and response when collision of information occurs) - <u>Flow control</u> (how much information is sent and its speed), - <u>Response timeout</u> (how long to wait for response).
		Message delivery options: - <i>Unicast</i> message (one-to-one) - <i>Multicast</i> message (one-to-many) - <i>Broadcast</i> message (one-to-all)
	Describe and outline key aspects of a protocol suite.	- It is a group of inter-related protocols that work together and are necessary to perform communication functions. - Protocols are viewed in terms of layers. (in a protocol stack) - Lower levels are concerned with <i>moving data over network</i> and <i>provision of services</i> to upper levels that focus on <i>content of message sent</i>.
	Outline network protocol suite and identify some examples.	- Describes precise requirements and interactions for devices to communicate with each other by defining common format and set of rules. - Examples: HTTP (Hypertext transfer protocol), TCP (Transmission Control Protocol), IP (Internet protocol).
	Describe the process of protocol interaction. (<i>HTTP, TCP, IP, Ethernet</i>)	HTTP: Application protocol, governs interaction between server and client. - Defines content and formatting of request/responses. - Relies on other protocols to govern transportation of messages. TCP: Transport protocol, manages individual conversation. - Divides HTTP messages into <i>segments</i>. - Size and rate of message exchanged are controlled by TCP. IP: Internet protocol, assigns appropriate address for delivery to destination host. - Encapsulates formatted <i>segments</i> from TCP into packets

Ethernet: Network access protocol. Responsible for formatting packets from IP to be transmitted over media.

Describes two primary functions:

- Communication over data link
- Physical transmission of data on network media

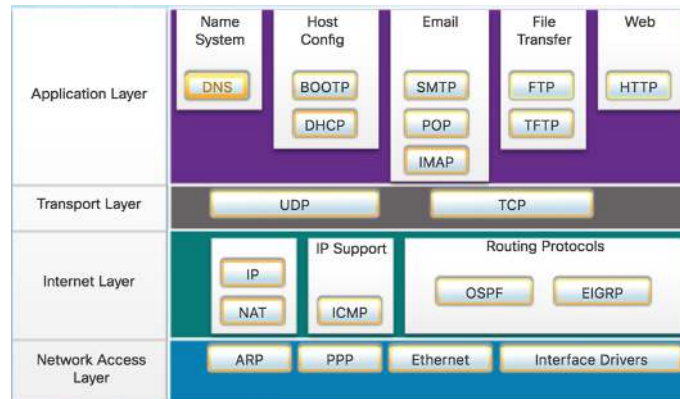
Discuss TCP/IP protocol suite (*Development, Examples*)

- Open standard (Freely available to public)

Development

- **ARPANET** (Advanced Research Projects Agency Network): 1st packet switching network and predecessor to internet today.

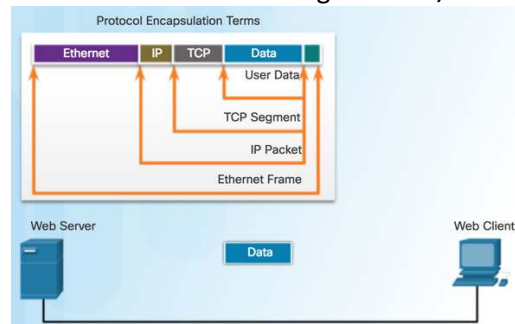
Example: protocols in TCP/IP protocol suite



Outline the TCP/IP communication process for **sending** AND **receiving** data between a client and webserver.

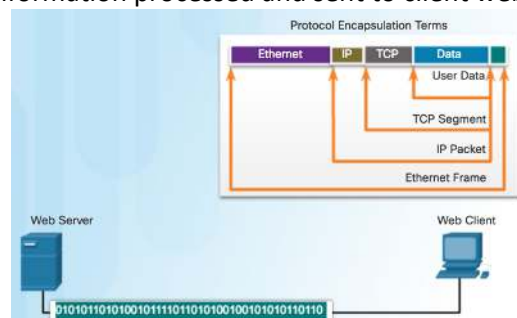
Sending data from web server to client

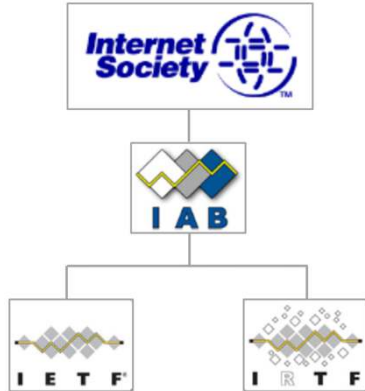
1. Webserver prepares HTTP page. HTTP application layer protocol *sends data to transport layer*.
2. Transport layer breaks data into *segments*.
3. *IP Packet* (IP sources & destination address) added by IP.
4. Ethernet information added creating *Ethernet/Data link frame*.



Receiving data link frames from webserver (remove each protocol layer in reverse to how it was added)

1. *Ethernet* header removed.
2. *IP* header removed.
3. *Transport* layer header removed.
4. *HTTP* information processed and sent to client web browser.



LECTURE 3: Network Protocols and Communication	Evaluate open standards.	- Open standards: being freely accessible to public - Encourages Inter-operability, competition and innovation - Prevents monopolization of product market
	What are standards-based protocols?	- <i>Process</i> that has been endorsed by the networking industry and approve by a standards organization. - Ensures products from different manufacturers can inter-operate successfully.
	Define proprietary protocols.	- A company or vendor <i>controls the definition</i> of the protocol and how it <i>functions</i> for the users. - E.g. AppleTalk, Novell Network (legacy protocol suites)
	Identify and outline roles of some Inter-Standards organisations.	- Standard organisations are: Vendor-neutral, non-profit, Promote concept of open standards.  <pre> graph TD IS[Internet Society] --- IAB[IAB] IAB --- IETF[IETF] IAB --- IRTF[IRTF] </pre> - ISOC: Responsible for promoting open development and evolution of internet use. - IAB: Responsible for overall management and development of internet standards - IETF: Develop (new protocols), update, maintain internet TCP/IP technologies. - IRTF: Focused on long term research related to internet and TCP/IP protocols - ICANN: Coordinates IP address allocation, management of domain names, - IANA: Responsible for overseeing and managing IP address allocation, domain name management.
	Identify advantages (4) of using a Layered model to describe network protocols and operations	1. Assisting in protocol <i>design</i> as each specific layer of protocols have defined information they act upon. 2. Fostering <i>competition</i> as products from different vendors can work together. 3. Technology and capability changes are <i>independent in layers</i>. 4. <i>Common language</i> to describe functions and capabilities.
	Explain the OSI (Open Systems Interconnection) Reference model . (<i>Used for network functionality</i>) (Essential to remember)	- Provides extensive list of functions and services at each level Application: Protocols used for process-to-process communication. Presentation: Provides for common representation of data. Session: Provide services to presentation layer (organize data exchange). Transport: Define services to: segment, transfer and reassemble. Network: Provides services to exchange individual pieces of data over network. Data link: Provides methods for exchanging data frames between devices over common media. Physical: Describes mechanical, electrical, functional, procedural means to transmit bits across.
	Explain the TCP/IP protocol model (also known as Internet Model). (<i>Used for network functionality</i>)	- Open standard. 1970s: for internetwork communication - <i>Four categories</i> necessary for successful communication. Application: Represents data to user (encoding +dialog control) Transport: Support communication between various devices Internet: Determine best path through network Network access: Controls hardware devices and media in network

LECTURE 3: Network Protocols and Communication	Discuss the OSI and TCP/IP model. <i>(Explain each layer's function and how they map to each other in the two models)</i> <i>(refer to sections individually explaining models)</i>		
	Explain the segmentation of messages and the benefits of doing so.	Message segmentation is the breaking of large streams of data into smaller and more manageable sizes for better reliability. - Allows missing segments of data that have failed to be received to be sent again rather than the entire stream of data. - Allows for multiplexing: Different conversations are interweaved	
	What is the Protocol data unit (PDU) that maps to each layer of the encapsulation process?	Data: Application layer PDU Segment: Transport layer PDU Packet: Network layer PDU Frame: Data link layer PDU Bits: Physical layer PDU	
	Explain de-encapsulation	The process of removing each protocol header, by the receiving device.	
	Identify purposes of the source and destination address of network and data-link layer. Also state the main difference/s between the two.	Network layer: Delivering <i>IP packet</i> from source to destination on same network or <i>remote network</i>.	Data link layer: Delivering <i>data link frame</i> from one network interface card to another on the <i>same network</i>.
		Difference: Network layer delivers IP packet whereas Data-link layer deliver data link frame. Network layer also delivers to remote networks	

LECTURE 4: Network Access	Identify the two types of connections that are used for network communication	Wireless: Data transmitted through radio waves. Devices on wireless network must be connected to WAP. Physical: Data transmitted through wired connection via cable
	Describe what NICs do. Outline	Network interface cards connect devices to a network - Used for wired connection - Negligible performance degradation - Separate communication channel each WLAN NICs used for wireless connection - Performance degradation over distance from WAP due to weak signal = less bandwidth - All wireless devices share 'airwaves' = Slower if many
	Explain the purpose of the physical layer	1. Provides means for transporting the bits that make up data link layer frame across network media. 2. It accepts complete frame from the data link layer and then encodes as series of signals. 3. Encoded bits received by end device/intermediate device
	Identify Physical layer medias (3).	Copper cable, fibre-optics, wireless. Refer to <i>Media</i> of components in network infrastructure (Lecture 1)
	Identify organisations involved in maintaining physical layer standards.	Physical layer includes: hardware, media, encoding, signalling Organisations: ISO, TIA/EIA, ITU, ANSI, IEEE, FCC, ETSI *Note: TCP/IP standards are implemented and governed by IETF. (Every layer above the physical layer, from data-link to application.
	What are the functional areas (3) which the physical layer addresses?	1. Physical connection: Hardware devices, media, NIC, interface etc. 2. Encoding: Converting streams of data bits into predefined code - 'Language' /pattern that both sender and receiver recognise - Manchester encoding: 0 bit = high → low voltage transition 3. Signalling: Defining what represents a "0" and "1" on electrical, optical and wireless signals.
	Explain what bandwidth is and factors that determine it.	- Definition: The capacity of a medium to carry data. - Measures the amount of data (in bits) that can flow /period. - b/s, kb/s, Mb/s, Gb/s, Tb/s Factors affecting bandwidth 1. Properties of physical media (physics) 2. Technology used for signalling and detecting network signals
	Explain what Throughput is and factors that determine it.	- The measure of the <i>transfer</i> of bits across media / period Factors affecting Throughput 1. Amount of traffic 2. Type of traffic 3. Latency created by travel of data *Note: Goodput is the application-level throughput. Measure of <i>usable</i> data transferred /period
	Outline characteristics of copper cabling and how interference can occur.	Used because: Inexpensive, easy installation, low resistance Disadvantages: Limited by distance and signal interference Interference occurs by: - EMI/RFI (radio frequency): distorts and corrupts data signals. This can be prevented by wrapping cables in metallic shielding and grounded. - Crosstalk: Disturbance caused by electric/magnetic fields produced signal on adjacent wire. This can be prevented by opposing circuit wire pairs twisted together to cancel out the crosstalk.

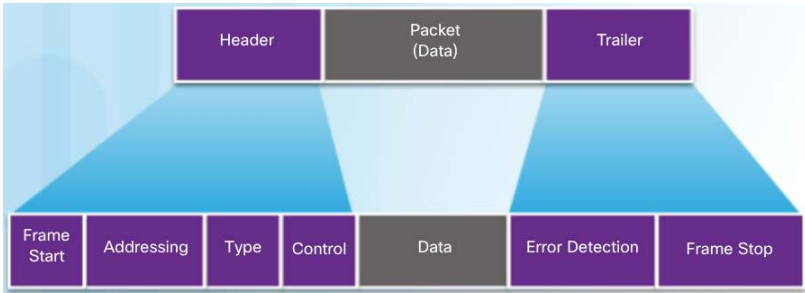
Identify how susceptibility of copper cables to electronic noise can be limited	1. Selecting cable type or category most suited to given networking environment 2. Designing cable infrastructure to avoid known and potential sources of interferences 3. Use cable techniques that include proper handling and termination of cables.
Identify the three main types of copper cabling. UTP  STP  Coaxial 	1. Unshielded Twisted Pair (UTP) - Most common used - Terminated with RJ-45 connectors - 4 pairs of colour-coded twisted wires (prevent signal interference), in plastic sheath (prevent physical damage) 2. Shielded Twisted Pair (STP) - Better noise protection than UTP but more expensive/difficult installation - Terminated with RJ045 connectors - Combines techniques of shielding to counter EMI and RFI and twisting to counter crosstalk. - 4 pairs of wires wrapped in foil shield each and over all metallic braid 3. Coaxial - Layers: Copper conductor, Flexible plastic insulation, insulating material as shield for conductor, Cable jacket (prevent damage) - Terminating connectors: BNC, N-type, F-type Used in Ethernet installation but now replaced by UTP
Copper media safety	- Fire hazard: Sheath is flammable/ toxic fumes when heated - Electrical hazard: Cables conduct in undesirable ways, voltage levels, damaging related components and devices - Prevention: Grounding properly, inspections, connections and comply with safety codes
Outline properties of UTP Cabling	- Consists: 4 pairs of colour-coded copper wires twisted together and encased in flexible plastic sheath - Doesn't use shielding to counter EMI and RFI. - Uses: Cancellation (Opposing B cancel out EMI and RFI signals), and vary number of twists per wire to enhance cancellation - Advantages: Small in size
Briefly describe each cable standard for UTP	- Standards established by TIA/EIA-568 for LAN installations 1. Cat 3 - Voice communication – Telephone lines 2. Cat 5 (100Mb/s-1000Mb/s) and 5e (1000Mb/s) (<i>minimum</i>) - Data transmission 3. Cat 6 (1000Mb/s – 10Gb/s) (<i>recommended</i>) - Data transmission - Has separation between each pair of wires for higher speeds
Identify types of UTP connectors	- Usually terminated with RJ-45 (One use is Ethernet) - Male component is inserted into Female component *Note: Badly terminated copper cabling can cause signal loss and noise.
Describe types of UTP cables	Ethernet Straight-through: Most common for networking (interconnect switch to host/router) Ethernet crossover: Interconnect similar devices (switch to switch) Rollover: Cisco proprietary (workstation to router/switch console)

LECTURE 4: Network Access	Identify the parameters to be tested after installation of UTP cable	Wire map, cable length, signal loss due to attenuation, cross talk.		
	Outline properties of Fibre-Optic Cabling	- Consists: Flexible thin strand of pure glass, data encoded as light pulses, fibre optic cable is 'light pipe' for transmission - Light is unidirectional so 2 fibres for duplex-operation - Advantages: Transmit data over long distances, Higher bandwidths, Less signal attenuation, immune to EMI and RFI		
	Outline types of Fibre-optic cabling services provided (4).	- Enterprise Networks: Backbone cabling application, interconnect infrastructure devices - Fibre to the Home (FTTH): Broadband services to homes - Long Haul networks: Servicer to connect country/city - Submarine cable networks: Reliable, high speed + capacity for transoceanic distances		
	Describe the design of Fibre Media cable	- Core: Transmits light pulses (made of silica or glass) - Cladding: Slightly different to core materials, Act like mirror to bounce light back into core - Buffer: Shields core and cladding from damage - Strengthening material: Prevents stretching of fibre cable - Jacket: Protect from abrasion, moisture (physical damage)		
	Outline types of Fibre Media	Single Mode (SMF) - 1 straight path for light - Small core - Less dispersion - Long distance - Light source: laser - Common use: Several thousand meters	Multimode (MMF) - Larger core - More dispersion = loss of signal - Long distance but shorter than single mode - Light source: LED - Common use: LANs, Several hundred meters	
	Identify connectors of Fibre optic connectors	Straight tip connector (ST): Locks securely with 'twist on/off' Subscriber connector (SC): Standard connector, Push pull mechanism, Both multimode/single mode fibre Lucent connector Simplex connector (LC): Smaller vs. of SC Duplex multimode LC connector: Similar to LC but duplex		
	Testing Fibre cables	Common termination and splicing errors: - Misalignment: Unaligned when joined - End gap: Doesn't not touch at splice or connection - End finish: Ends not polished/ dirty Tested by: Bright flash light through one end and observe the other		
	Compare and contrast Fibre and copper cabling	Implementation Issues	UTP Cable	Fibre-optic cable
		Bandwidth support Distance EMR/RFI Immunity Elec. hazard immunity Cost Installation skill Safety precautions	10Mb/s – 10Gb/s Short (100m) Low Low Low Low Low	10Mb/s – 100Gb/s High (100,000m) 100% immune 100% immune High High High
	Outline properties of wireless media	- Electromagnetic signals to carry data: radio/microwave - Most mobility of all media types hence popularity - Concerns: coverage area (buildings), interference (other waves of devices on same frequency), security (don't need access to physical media), shared medium (more user = less bandwidth)		

LECTURE 4: Network Access

Identify types of wireless media	- Wi-Fi: Uses CSMA/CA. Wireless NIC waits until channel is clear. - Bluetooth: Wireless Personal Area Network. Uses pairing process (1-100m) - Wi-MAX: Worldwide Interoperability for Microwave Access. Wireless broadband Access	
Describe a wireless LAN and its requirements	- Enables devices to connect wirelessly via a LAN. - This requires: 1. Wireless Access point: Concentrate signals and connects to existing network infrastructure 2. Wireless NIC adapter: Provide wireless communication capability to each network host - Wireless data communications save costs on wiring and enables mobility. But cost for preventing unauthorized access.	
Outline the purpose of the Data Link layer (Layer 2 of OSI model)	Layer 2 notation for network devices connected to common media is called a node. This builds and forwards frames. Data link layer separates media transitions that occur as packet is forwarded from communication process of higher layers. Data link layer is also responsible for: - Allow upper layers to access the media - Accept Layer 3 packets and packaging them into frames - Preparing network data for physical network - Control data placement and how it is received on the media - Exchanging frames over nodes over physical network media - Receiving and directing packets to an upper layer protocol - Performing error detection	
Describe the two sublayers of data link	Logical Link Control (LLC) - Upper sublayer - Communicates with network layer - Play information in frame that identifies network layer protocol is being used for the frame.	Media Access Control (MAC) - Lower sublayer - Defines media access processes performed by hardware. - Provides data link layer addressing and access to network technologies
Outline the concept of Media Access control	Media access control method: Technique for getting frame on and off media. (Layer 2 is the encapsulation of packet into a frame) - Travelling from source host to destination host, packets transverse over different physical networks (p-media types). - Without data link layer, network layer protocols (e.g. IP) have to make provisions for connecting every type of media in the delivery path.	
Briefly outline the steps that media access is provided for different physical networks.	At each hop on the delivery path: 1. Router accepts the frame from medium 2. De-encapsulates the frame 3. Re-encapsulates packet into a new frame 4. Forward new frame to medium of that segment of the physical network	
Identify data link layer standards	Defines open standards and protocols that ally to network access layer: IEEE, ITU, ISO, ANSI	
Explain controlling the access to the media	Analogy: Traffic rules that regulate the entrance of motor vehicles onto a road. Absence of this means vehicles ignore traffic. Also, not all roads and entrances are the same (different rules). The MAC method used depends on: Topology & Media sharing	

LECTURE 4: Network Access	Explain Logical and physical topologies	Physical - Physical connections and how devices are interconnected.	Logical - Method which network transfers frames from one node to the next. - Defined by data link layer protocols
	Identify common (3) physical WAN topologies	- Point-to-point: Permanent link between two end points - Hub and spoke: Central site interconnecting branch sites using point-to-point - Mesh: Interconnects every end to every other system. High availability and costs.	
	Explain physical and logical Point-to-Point topology	Physical - Directly connects two nodes - Doesn't not share the media with any other hosts Logical - End nodes of point-to-point network can be connected via many intermediate devices - Logical connection between nodes is called virtual circuit - Media access method used by data link protocol is determined by logical topology	
	Explain Physical LAN topologies	"How end systems are physically interconnected" - Star – End devices connected to central intermediate device using ethernet switches - Extended star – Additional ethernet switches inter connect other star topologies - Bus – Legacy Networks. All end system chain to each other. Uses coax cables as inexpensive and easy setup. - Ring – End systems connected to respective neighbour forming a ring. No termination needed. Used in FDDI and Token ring networks.	
	Describe half and full duplex communication	Duplex communication: Direction of data transmission between two devices. - Half Duplex: Both devices can transmit and receive on media but not simultaneously. Used in legacy bus topologies, WLAN. Used with contention-based access methods. - Full Duplex: Both devices can transmit and receive on the media at the same time. Used by Ethernet switches (can also use half duplex)	
	Identify several access control methods for shared media (ethernet LANs/ WLANs)	Contention-based access: - All nodes operating in half-duplex compete for use of medium Controlled access: - Each node has its own time to use medium.	
	Describe CSMA/CD (Contention-based access)	<u>Carrier Sense Multiple Access/ Collision Detection Process</u> - If two devices transmit at the same time, collision will occur. - Devices will detect collision on network through NICE data comparison of data transmitted and data received. - The data will be corrupted and needs to be resent.	
	Describe CSMA/CA (Contention-based access)	<u>Carrier Sense Multiple Access/ Collision Avoidance Process</u> - Similar to CSMA/CD to detect if media is clear. - Doesn't not detect collision but attempts to avoid by waiting before transmission - Calculates time duration needs for other transmission and other devices will wait until medium becomes available again.	

LECTURE 4: Network Access	The Frame and its fields	Data link layer prepares packet of data for transport by encapsulating it with header and trailer (makes a frame). Frame Fields  Frame start/stop: Identify start and end of frame. Addressing: Source and destination nodes Type: The Layer 3 protocol in data field Control: Identify special flow control services (QoS) Data: Frame payload
	LAN and WAN frames	Layer 2 protocols used for a topology is determined by technology Data link layer protocols include: Ethernet, 802.11 Wireless, PPP (Point to Point Protocol), HDLC, Frame Relay

LECTURE 5: Ethernet	Outline Ethernet Encapsulation	- Ethernet defines both layer 1 (physical) technologies and layer 2 (data link) protocols. Defined in IEEE 802.2/3 standards - Ethernet relies on two separate sublayers of data link (layer 2) to operate: LLC and MAC sublayers - LLC sublayer: Handles communication between upper and lower layers. Implemented in software and is independent of hardware. - MAC sublayer: Implemented by hardware and typically in the computer NIC.
	Describe and explain responsibilities of the MAC sublayer	1. Data encapsulation - Frame assembly and disassembly before and after transmission and reception. - When forming frame, MAC layer adds header and trailer to network layer PDU - Data encapsulation provides 3 primary functions: Frame delimiting, Addressing, Error detection (refer to Lecture 4 end) 2. Media access control - Responsible for placement and removal of frames on media - This sublayer communicates directly with physical layer
	Outline the evolution of Ethernet	- Created in 1973 (ethernet) - Early versions were slow at 10Mbps and now at 10Gbps
	Ethernet frame fields	*Refer to Lecture 4 end - Minimum Ethernet frame size from Destination MAC address to FCS is 64 bytes and max of 1518 bytes. - Collision fragment/runt frame- If less than minimum it is discarded by receiver. - Jumbo/baby giant frames – more than 1500 bytes.
	MAC address and Hexadecimal	- An Ethernet MAC address is a 48-bit binary value expressed as 12 hexadecimal digits (4 bits per hexadecimal digit). - Uses Hexadecimal system (base 16 system) (0 to 9 and A to F) - Easier to express a single hexadecimal than 4 binary digits - Hexadecimal is used to represent Ethernet MAC addresses and IPv6 addresses
	Ethernet MAC Address structure	Originally ethernet was half-duplex topology hence all nodes receive every frame transmitted. - MAC Addresses created to identify actual source and destination. - Established by IEEE to assign 3-byte (24 bit) code OUI Rule: - All MAC addresses assigned to a NIC or ethernet device must use the vendors assigned OUI as 1st 3-bytes - All MAC addresses with the same OUI must be assigned a unique value in last 3-bytes
	Outline process of frame forwarding	- MAC address also known as BIA (Burned in address). - Encoded into ROM chip permanently so that after start-up, ROM copies MAC address into RAM. - When forwarding message to ethernet network, it attaches a header (contains source and destination MAC address) - All devices that can be source or destination of an ethernet frame must be assigned a MAC address or BIA - ipconfig /all to identify MAC address

LECTURE 5: Ethernet	Unicast MAC address	- Different MAC addresses are used for layer 2 unicast, broadcast, and multicast communications - A unicast MAC address is the unique address used when a frame is sent from a single transmitting device to a single destination device. - To determine destination MAC address, Address Resolution Protocol is used (ARP) - Destination MAC access can be a unicast, broadcast, multicast but source must be unicast
	Broadcast MAC address	- Broadcast packet contains a destination IPv4 address with all '1' in host portion. - '1' means all host on local network (broadcast domain) will receive and process the packet. - E.g. DHCP and ARP use broadcasts - Destination MAC address is FF-FF-FF-FF-FF-FF
	Multicast MAC addresses	- A source device can send a packet to a group of devices. - Devices belonging to multicast group have multicast group IP address. (IPv4 range: 224.0.0.0 to 239.255.255.255) (IPv6 start with: FF00::/8)
	MAC Address table	- Switch learns and builds MAC address table through process: 1. Exam all incoming frames for new source MAC address info 2. If source MAC address it is added to table with port no. 3. If source MAC address exists, switch updates refresh timer 4. By default, most ethernet switches keep entry for 5 mins. - Switch forwards Destination MAC address: 1. If destination MAC is broadcast/multicast, the frame is also flooded out of all ports exam incoming port 2. If destination MAC address is unicast, switch will look for match in MAC address table 3. If destination MAC address is in table, it will forward frame out of specified port 4. If destination MAC address is unknown, switch will forward the frame out of all ports except incoming port.
	Frame forwarding methods (cisco devices)	Store and forward switching - When the switch receives the frame, it stores the data in buffers until the complete frame has been received. During the storage process, the switch analyses the frame for information about its destination. In this process, the switch also performs an error check using the Cyclic Redundancy Check (CRC) trailer portion of the Ethernet frame. If valid, destination address is found and used to determine outgoing interface. Cut through switching - Frame can be forwarded before it is entirely received. Destination address of frame must be read before forwarded. - No error checking is done on frame.
	Cut through switching types (2)	Fast forward switching - Lowest level of latency as packet is forwarded immediately after destination address is read. Most common Fragment free switching - Stores first 64-bytes of frame before forwarding. Compromise between the two types of forwarding methods. (in above)

LECTURE 5: Ethernet	Memory buffering	Technique to store frames before forwarding them. Can also be used when destination port is congested. - Port-based memory – In port-based memory buffering, frames are stored in queues that are linked to specific incoming and outgoing ports. Frame is transmitted to the outgoing port only when all the frames ahead of it in the queue have been successfully transmitted. - Shared memory – Shared memory buffering deposits all frames into a common memory buffer that all the ports on the switch share.
	Duplex and speed setting on switch port	- Two types of duplex communication: Half and full duplex. - Uses Auto-negotiation to enable two devices to exchange information about speed and duplex capabilities allowing switch to choose highest performance mode. - Duplex mismatch: One port on link operates at half duplex while other is at full duplex. (either both sides have auto-negotiation on or off)
	Auto-MDIX	- Connection between specific devices need to use specific cable types (p-media type) - Automatic medium-dependant interface crossover (mdix auto) to <i>detect type of cable used</i> and configures interface accordingly
	Destination on the same network	- Ethernet LAN device has: Physical (MAC) and Logical (IP) address - Layer 2: Ethernet frame has source and destination MAC address. Delivery data link frame with IP packet from NIC to another NIC - Layer 3: IP address of original source and destination address
	Address Resolution Protocol	- To determine destination MAC address, device uses ARP. Function: Resolve IPv4 address to MAC addresses - Refers to ARP table to <i>find the MAC address that is mapped to IPv4 address</i>. - If same network, device search ARP table for destination IPv4 address. If different network, device search ARP table for IPv4 address of default gateway. - Maintaining a table of mappings (ARP Table) - ARP cache timer removes ARP entries that have not been used for a specified time period. Can also manually remove. - To view, use show ip arp or arp -a depending on device
	ARP broadcasts	- An ARP request is received and processed by every device on the local network. Normally minimal impact. - Large number of powered up and access at same time can cause performance reduction.
	ARP spoofing	- Security risk of ARP spoofing or poisoning - Technique used to reply to an ARP request for IPv4 address of another device. Attacked will receive this information instead. - Enterprise level switches include DAI (Dynamic ARP inspection) to mitigate this.

LECTURE 6: Network Layer	Identify processes (4) of the network Layer (OSI Layer 3)	Provide services for end devices to exchange data across network 1. Addressing end devices - End devices must be configured with unique IP address to be identified in a network. 2. Encapsulation - Encapsulates protocol data unit (PDU) from transport layer into a packet. Encapsulation adds IP header. 3. Routing - Directing packets to a destination host on another network. Router will select best path and direct packets to other networks to the destination host through <i>routing</i>. - Packet may also cross intermediary devices. - Each router crossed to reach destination host is called a <i>hop</i>. 4. Decapsulation - Host checks IP header of packet, and if matching, header is removed and de-encapsulated.
	Protocols of the Network layer (2)	IPv4 and IPv6. <i>Legacy protocols are not in the course.</i>
	Explain IP encapsulation and its use	- IP <i>encapsulates</i> the transport layer segment or other data by adding IP header. (Doesn't change from source to destination) - Encapsulation of data layer by layer allows scalability of different layer services without affecting other layers. - Routers can implement these different network layer protocols to operate concurrently over a network
	Outline characteristics of IP	IP only provides functions needed to deliver a packet from source to a destination. (Media Independent) - IP is independent of media type (copper, optic fibre, wireless) as OSI data link layer takes the IP and prepares it for transmission over the communication medium detected) - Network layer considers the MTU (max. transmission unit). Data link passes the MTU value up to network layer to determine size of packets. IP packet is sent to destination without prior establishment with of a connection with the destination (Connectionless) - No end-to-end connection is created before data sent. - Sender does not know if destination is present, reachable or functional - Receiver does not know when it is coming *Hence the low overhead of IP Does not track and manage the flow of packets (Best effort) - Does not guarantee all packets that are delivered, are received - IP is not capable of managing/recovering corrupt packets as there is information to inform sender if delivery was successful - If packets are missing or not in order, upper layer protocols resolve these issues.
	IPv4 Packet header fields	Version: 4-bit binary value (IPv4 is 0100) Differentiated Services (DS): 8-bit field to determine priority of each packet. 6 most significant bits is DS code point and last two are Explicit congestion notification (ECN). Time to Live (TTL): 8-bit binary value to limit lifetime of packet. Protocol: Identifies next level protocol. 8-bit binary value to indicate data payload type to pass data to appropriate upper layer protocol

LECTURE 6: Network Layer

	Source IPv4 address: 32-bit binary value to represent the source IPv4 packet. Always Unicast. Destination IPv4 address: 32-bit binary value to represent the destination IPv4 packet. Can be unicast, broadcast, multicast. - Internet Header length, Total length, header checksum fields used to identify and validate packet																																																		
Limitations of IPv4 (3)	1. IP address depletion - Limited unique public IPv4 address available 2. Internet routing table expansion - Table made by routed for best path determination. As servers increase, no. of routes increases. IPv4 routes hence consume large amount of memory and processor resources. 3. Lack of end to end connectivity - NAT allows multiple devices to share 1 single public IPv4 address. IPv4 address of internal network host is hidden hence problematic for technologies that require end to end																																																		
IPv6 improvements in comparison to IPv4	IETF developed IPv6 due to issues of IPv4. 1. Increased address space – 128-bit hierarchical addressing (efficient) whereas IPv4 is 32-bit. And Autoconfiguration 2. Improved packet handling – simplified header with less fields, hence larger payload for throughput and transport efficiency. 3. Eliminates need for NAT – larger number of available IPv6 addresses.																																																		
Comparison of IPv4 and IPv6 header	IPv4 Header. (Grey is not in IPv6, Green has different name and position, Purple is same in IPv6) <table><tr><td>Version</td><td>IHL</td><td>Type of Service</td><td colspan="2">Total Length</td></tr><tr><td colspan="2">Identification</td><td>Flags</td><td colspan="2">Fragment Offset</td></tr><tr><td>Time-to-Live</td><td>Protocol</td><td colspan="3">Header Checksum</td></tr><tr><td colspan="5">Source Address</td></tr><tr><td colspan="5">Destination Address</td></tr><tr><td colspan="3">Options</td><td colspan="2">Padding</td></tr></table> IPv6 Header (Purple is same field, Green is different name and position, <i>Blue is a new field</i>) <table><tr><td>Version</td><td>Traffic Class</td><td colspan="3">Flow Label</td></tr><tr><td colspan="2">Payload Length</td><td>Next Header</td><td colspan="2">Hop Limit</td></tr><tr><td colspan="5">Source IP Address</td></tr><tr><td colspan="5">Destination IP Address</td></tr></table>	Version	IHL	Type of Service	Total Length		Identification		Flags	Fragment Offset		Time-to-Live	Protocol	Header Checksum			Source Address					Destination Address					Options			Padding		Version	Traffic Class	Flow Label			Payload Length		Next Header	Hop Limit		Source IP Address					Destination IP Address				
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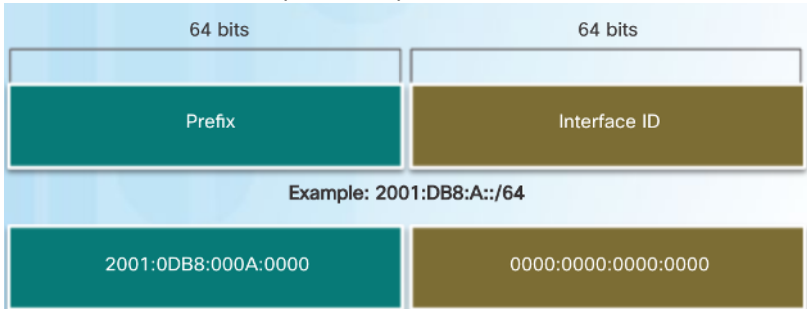
LECTURE 6: Network Layer	IPv6 packet header fields	Version: 4-bit binary value (IPv4 is 0110) Traffic class: 8-bit field equivalent to DS field of IPv4. Flow label: 20-bit field suggesting all packets with same flow label receive same type of handling by router Payload length: 16-bit field to indicate length of the data payload of the packet. Next header: 8-bit field equivalent to the IPv4 protocol field. Hop limit: 8-bit field replacing IPv4 TTL field. Value is subtracted by 1 for each router it passes. Discarded when 0. Source IPv6 address: 128-bit field to represent the source IPv6 host Destination IPv6 address: 128-bit field to represent the receiving IPv6 host.
	Directing packets between hosts	A host can send a packet to 1. Itself: Ping itself for testing purposes (loopback interface) 2. Local host: Same network, hence same network address 3. Remote host: remote network, hence different address Source IPv4 and subnet mask is compared with destination to determine if host is local or remote. If remote, router is needed to connect to local network segment called default gateway
	Explain default gateway and its use	- A network device that can route traffic to other networks. - Has local IP address in the same range as other hosts on same network - Can take in and forward data
	Directing packets by routers	- Host A >> Router 1 (default gateway) >> Router 2 (default gateway) >> Host B Routing table stores information about: - <i>Directly connected routes:</i> Active router interfaces - <i>Remote routes:</i> Remote networks - <i>Default route:</i> Last resort if no other route
	Directly connect routing table entries	- When router interface with IPv4 address, subnet mask is activated, two entries are made automatically C (identifies directly connected network) and L (identifies this is a local interface)
	Remote network routing table entries	- Route source – How network was learnt of by router (S, D, O) - Destination network - Administrative distance – trustworthiness of route source - Metric – Assigned value to reach remote network - Next-Hop – Next IP address of next router to forward to (C and L doesn't not have next hop address as it is direct) - Route timestamp – When router was last heard from - Outgoing interface – Interface used to forward packet
	Describe a router	- Types: Branch (small to medium size), WAN (large business/enterprises), Service provider Consists of: CPU, OS (usually Cisco Internetwork OS/IOS), Memory (RAM, ROM, NVRAM, flash)
	Router memory types	- Volatile: continue power to store information - Non-volatile: doesn't need continual power RAM – Volatile for application, processes, data executed by CPU ROM – Non-volatile for operation instruction and IOS. (firmware) NVRAM – Non-volatile for permanent storage of startup-config Flash – Non-volatile for permanent storage for IOS, log, backup files

LECTURE 6: Network Layer	Components of a router	- Generally, routers all have the same hardware components with little variation - Power supply, Cooling fan, Synchronous Dynamic RAM, NVRAM, CPU, Heat shield, Advanced Integration module
	Ports and connections on a router	- eHWIC: Enhanced High-speed WAN interface card - Auxiliary (AUX): remote management - Console port: Configuration - Gigabit Ethernet: LAN access connections - Compact flash slots: Increase storage - USB port: additional storage space
	Accessing LAN and WAN interfaces of a router	- Connections classify into two types: In-band interfaces and Management ports - Accessing user EXEC mode in CLI: Console, SSH, Telnet
	Router bootup process	- Perform POST and load bootstrap program - Locate, load cisco IOS - Locate, load start-up config

LECTURE 7: IP addressing

Describe IPv4 Addresses in basic terms.	- 32 binary bits divided into 4 x 8-bit octets - Dotted decimal notation																																				
Explain Binary Positional notation	<table><tr><td>Radix</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td></tr><tr><td>Position in Number</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>Calculate</td><td>(2^7)</td><td>(2^6)</td><td>(2^5)</td><td>(2^4)</td><td>(2^3)</td><td>(2^2)</td><td>(2^1)</td><td>(2^0)</td></tr><tr><td>Positional Value</td><td>128</td><td>64</td><td>32</td><td>16</td><td>8</td><td>4</td><td>2</td><td>1</td></tr></table> Total positional value is 192.	Radix	2	2	2	2	2	2	2	2	Position in Number	7	6	5	4	3	2	1	0	Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)	Positional Value	128	64	32	16	8	4	2	1
Radix	2	2	2	2	2	2	2	2																													
Position in Number	7	6	5	4	3	2	1	0																													
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)																													
Positional Value	128	64	32	16	8	4	2	1																													
Describe Network and Host portions	- IPv4 is hierarchal (2 portions) IPv4 Address <table><tr><td colspan="3">Network Portion</td><td>Host Portion</td></tr><tr><td>192</td><td>168</td><td>10</td><td>10</td></tr><tr><td colspan="3">11000000 10101000 00001010</td><td>00001010</td></tr></table> - Devices on same network have same network portion - Subnet mask identifies network portion and host portion of devices	Network Portion			Host Portion	192	168	10	10	11000000 10101000 00001010			00001010																								
Network Portion			Host Portion																																		
192	168	10	10																																		
11000000 10101000 00001010			00001010																																		
Describe a Subnet mask	- Used to determine network address where device belongs - A '1' in the subnet mark indicates that the corresponding bit in the IPv4 address is a network portion bit. - Identifying the network and host portion is called ANDing																																				
Explain what Logical AND is	- One of the 3 basic binary operations (others are OR, NOT) - Corresponding binary '1' bits in both the IPv4 and Subnet are also listed as '1' in the AND results. The remaining are set as '0' - 1 & 1 = 1 0 & 1 = 0 1 & 0 = 0 0 & 0 = 0																																				
Outline what a Prefix length is	- No. of bits in the subnet mask that are set to '1' - Represents subnet mask																																				
Explain the types of addresses in a network using an example	- Example: 192.168.10.0/24 - Network address: host portion is all '0's. (by ANDing) - First host address: host portion is all '0's and ends with '1' - Last host address: host portion is all '1's and ends with '0' - Broadcast address: host portion is all '1's																																				
Explain Static IPv4 address assignment	- Devices like printers, servers and networking devices do not change IP address hence a static IP address will be assigned.																																				
Explain Dynamic IPv4 address assignment	- Impractical to assign static IPv4 address for devices like PC, phones, tablets, etc. Hence, uses DHCP (Dynamic host configuration protocol). - Requests IPv4 address from DHCP server. IPv4 address is 'leased' for a period of time until it is taken off network																																				

LECTURE 7: IP addressing	IPv4 communication types	Unicast – One to one Broadcast – One to all - Message sent to everyone in broadcast domain - Destination IPv4 address has all '1' in host portion Multicast – One to select group - Routing protocols use multicast transmission to exchange routing information - 224.0.0.0 to 239.255.255.255 reserved for multicast	
	Public and private IPv4 addresses	Public - Globally routed by ISP - Not all available IPv4 addresses can be used	Private - Not routable - Internal networks only - Must be translated to public IPv4 to be routable (using NAT)
	Identify Special User IPv4 addresses	Loop back addresses - To direct traffic to itself for testing if TCP/IP config is operational. Link local addresses - Also known as Automatic Private IP addressing. Windows DHCP client self-configure when no DHCP servers available. Used in Peer to peer. TEST-NET addresses - Addresses are set for teaching and learning purposes.	
	Legacy Classful Addressing	Network addressing based on 3 classes in 1981 - Class A: 16 mil hosts - Class B: 65000 hosts - Class C: 254 hosts Classful addressing wasted addresses and exhausted IPv4 availability	
	Classless addressing	- Classless introduced in 1990's - CIDR: allow service provider to allocate IPv4 addresses on any bit boundary instead of only class A,B,C - CIDR was temporary solution and new IP protocol needed to be developed for rapid growth of internet users. (IPv6)	
	Who assigns IP addresses	- ARIN: North America - RIPE: Europe, middle east, central Asia - APNIC: Asia and pacific regions - AfriNIC: Africa - LACNIC: Latin America and Caribbean islands	
	Identify need for IPv6	Advantage over IPv4: 128bit vs 32 bit - 340 undecillion addresses available - Solves limitations of IPv4 - Auto configuration enhancement Why is it needed: - Rapid increasing internet users - Depletion of IPv4 addresses - NAT issues - Internet of Things	

LECTURE 7: IP addressing	IPv4 coexistence with IPv6	- IETF created various protocols and tools to migrate networks to IPv6 - Dual stack: IPv4 and IPv6 on same network segment. Devices will run both IPv4 and IPv6 protocols simultaneously - Tunnelling: Transporting IPv6 packet over IPv4 network. IPv6 encapsulated inside IPv4 packet. - Translation: NAT64 allows IPv6 to communicate with IPv4 devices similar to NAT.
	Representation of IPv6 addresses and rules	- 128 bits in length as string of hexadecimal values (32 x 4) - Hextet: 16 bits or four hexadecimal values - Preferred format: X : X : X : X : X : X : X : X Rule 1: Omit leading 0s (Any '0's in front of letters or last '0') Rule 2: Omit all 0 segments (use '::' (double colon) when a hextet consists of only '0's. Can only be used once)
	Identify types of IPv6 addresses	- Unicast: Uniquely identifies an IPv6 interface device. Source IPv6 is always unicast address. - Multicast: IPv6 address used to sent a single IPv6 packet to many locations. - Anycast: IPv6 unicast address that can be assigned to many devices. Packet is routed to nearest device with that address. * IPv6 doesn't have IPv4's broadcast, but IPv6 'all-nodes'
	Describe the IPv6 prefix length	- Indicates network portion of an address (other part is an interface ID/host portion) - Most LANs is /64 (1 to 128)  The diagram illustrates the structure of an IPv6 address. It shows a 128-bit address divided into two 64-bit parts: the Prefix and the Interface ID. An example is provided: 2001:0DB8:A::/64. The prefix part is 2001:0DB8:000A:0000 and the interface ID part is 0000:0000:0000:0000.
	IPv6 unicast addresses	- Global unicast: Globally unique, internet routable addresses. Can be static or dynamic. - Link-local: Communicate with other devices on same local link (or subnet). Confined to a single link. Non-routable beyond that link. - Unique Local: Local addressing within a site or between limited no. of sites. Non-routable in global IPv6.
	Outline structure of an IPv6 Global unicast address	3 parts - Global routing prefix: Assigned by provider. Typically /48 - Subnet ID: Used to subnet in an organisation - Interface ID: IPv4 host portion equivalent. 1st four hextets are network portion, 4th hextet is subnet ID, last 4 are Interface ID.

LECTURE 7: IP addressing	Dynamic configuration of IPv6 addresses	SLAAC: Stateless Address Autoconfiguration - Devices obtain information from IPv6 router. - Uses ICMPv6 Router Advertisement messages to get messages it needs. (every 200 sec.) - RA message reply is called ICMPv6 Router Solicitation ICMPv6 RA message consists (3) of: - Network prefix and length - Default gateway address - DNS addresses and domain name Options for RA messages (3): 1. SLAAC 2. SLAAC with stateless DHCPv6 server 3. Stateful DHCPv6 (no SLAAC) - SLAAC for IPv6 global unicast address and default gateway - Uses stateless DHCPv6 for all others - DHCPv6 for all others
	EUI-64 process or random generation	- When option 1 or option 2, client must generate own Interface ID. (using EUI-64 or random generated 64-bit no.) - EUI interface ID is binary with 3 parts: 1. 24-bit OUI from mac address with 7th bit reversed 2. Inserted 16-bit FFFE in hexadecimal 3. 24-bit device identifier from MAC address
	Types of IPv6 multicast addresses (2)	- Assigned multicast: Reserved addresses for predefined groups of devices - Solicited node multicast: Mapped to a special Ethernet multicast address. Ethernet NIC filter frame on destination MAC. Common assigned groups: FF02::1 (all nodes, similar to broadcast in IPv4) and FF02::2 (all routers)
	What is ICMPv4 and ICMPv6	- Messaging protocols for IPv4 and IPv6 respectively - Both include: Host confirmation, Destination or service unreachable, Time exceeded, Route redirection
	ICMP v6 RS and RA	- Router Solicitation and Router Advertisement Messages Messaging between IPv6 router and IPv6 device - RS, and RA Messaging between IPv6 devices - NS, and NA NS, NA used for Address resolution and Duplicate address detection.

LECTURE 8: Subnetting IP networks

Broadcasting in an Ethernet LAN (broadcast domain)	For locating: - Other devices: ARP sent layer 2 broadcast to IPv4 address to find the associated MAC address - Services: DHCP sent broadcast to find DHCP server Switch propagate broadcast out of all interface except interface it was received Routers do not propagate broadcasts																																					
Large broadcast domains	- Network that connects many hosts - Host can generate excessive broadcast and affect network - Slows network operations due to traffic - Slow device operative due to a device accepting and processing each broadcast packet - Solved: Subnetting (smaller network spaces inside broadcast domains)																																					
Why do we use subnetting\	- Reduce overall network traffic and improve performance - Enable security policies between subnets																																					
Subnetting on Octet boundary	- Prefix length and subnet mask used to identify network portion - Subnet created by using host bits as network bits - More bits borrowed = more subnets = less hosts																																					
Classless Subnetting	<table><tr><th>Prefix Length</th><th>Subnet Mask</th><th>Subnet Mask in Binary (n = network, h = host)</th><th># of subnets</th><th># of hosts</th></tr><tr><td>/25</td><td>255.255.255.128</td><td>nnnnnnnn.nnnnnnnn.nnnnnnnn.nh h h h h h h 11111111.11111111.11111111.10000000</td><td>2</td><td>126</td></tr><tr><td>/26</td><td>255.255.255.192</td><td>nnnnnnnn.nnnnnnnn.nnnnnnnn.nn h h h h h h 11111111.11111111.11111111.11000000</td><td>4</td><td>62</td></tr><tr><td>/27</td><td>255.255.255.224</td><td>nnnnnnnn.nnnnnnnn.nnnnnnnn.nnn h h h h h h 11111111.11111111.11111111.11100000</td><td>8</td><td>30</td></tr><tr><td>/28</td><td>255.255.255.240</td><td>nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnn h h h h 11111111.11111111.11111111.11110000</td><td>16</td><td>14</td></tr><tr><td>/29</td><td>255.255.255.248</td><td>nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnn h h h 11111111.11111111.11111111.11111000</td><td>32</td><td>6</td></tr><tr><td>/30</td><td>255.255.255.252</td><td>nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnn h h 11111111.11111111.11111111.11111100</td><td>64</td><td>2</td></tr></table>			Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts	/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.nh h h h h h h 11111111.11111111.11111111.10000000	2	126	/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.nn h h h h h h 11111111.11111111.11111111.11000000	4	62	/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnn h h h h h h 11111111.11111111.11111111.11100000	8	30	/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnn h h h h 11111111.11111111.11111111.11110000	16	14	/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnn h h h 11111111.11111111.11111111.11111000	32	6	/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnn h h 11111111.11111111.11111111.11111100	64	2
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Subnetting formulas	2^n n = bits borrowed No. of subnets		2^{n-2} n = the number of bits remaining in the host field No. of Hosts																																			
Host Requirements when subnetting	1. Consider number of hosts addressed needed per network 2. Consider number of individual subnets needed																																					
Traditional subnetting	- Divides address space into adequate no. of subnets - Wastes a lot of unused addresses - Solved: Prevent waste of address using VLSM (Variable length subnet mask)																																					
VLSM	- Subnet size doesn't need to be equal																																					
Address allocation planning	3 Considerations 1. Prevent duplication of address (all unique) 2. Provide and control access 3. Monitor security and performance of host (Traffic examined for IP address that is generating excessive)																																					

LECTURE 8: Subnetting IP networks	Assigning address to devices	End user clients - Allocate address using DHCP to reduce burden. - Address are only leased for period of time. Servers and peripherals - Predictable static IP address used Intermediary devices - Assigned addresses for management, monitoring and security. - Static, predictable addresses Gateway - IP address assigned to interface that acts as gateway for hosts in that network. (Usually lowest/highest address)
	IPv6 Global Unicast	IPv6 address plan can focus on best hierarchical approach to manage and assign IPv6 subnets 2-types of assignable IPv6 addresses 1. Link local address - Never subnetted as it only exists on local link 2. IPv6 global unicast - Normally /48 global routing prefix, 16 bit subnet ID, 64 bit interface ID

LECTURE 9: Transport Layer	Role of transport layer (layer 4 OSI)	- Establish temporary communication session between two applications and delivering data between them - Link between application layer and the lower layers (for network transmission)	
	Transport layer responsibilities (3)	1. Tracking the conversations 2. Segmentation 3. Identifying application - Track each individual conversation between source and application destination - Divide data into segments for easier management and transportation. Transport layer header added for reassembly. - Ensure all applications receive correct data via port no.	
	Conversion multiplexing	Essentially segmenting data into smaller chunks to enable different communications to be multiplexed on the same network Header will be added to manage this data communication	
	Reliability of the transport layer	- Transport layer is responsible for managing reliability requirements of conversations - Transport layer specifies how to transfer messages between hosts through protocols: - o Transmission control protocol (TCP) Considered reliable. Additional fields need in header - o User datagram protocol (UDP) Doesn't provide for reliability. Few fields = faster than TCP	
	Operations of reliability for TCP (3)	TCP transport analogy: Send package that is tracked from source to destination. Shipping order is broken up into several packages. 1. Numbering and tracking data segments 2. Acknowledging received data 3. Retransmitting unacknowledged data after certain time	
	UDP compared to TCP	- Basic function for delivering data segments 1. Best effort delivery protocol (unreliable) 2. No acknowledgement	
	Right transport layer protocol to use for an application	UDP - Fast - Low overhead - Doesn't need acknowledgement - Delivers data as it arrives	TCP - Reliable - Acknowledges - Resends lost data - Deliver data in sequenced order
		Uses: IP Telephony, Streaming live video	Uses: Email (SMTP/POP), HTTP
	TCP Features	Establishing a session - TCP is connection-oriented protocol - Permanent link/session	

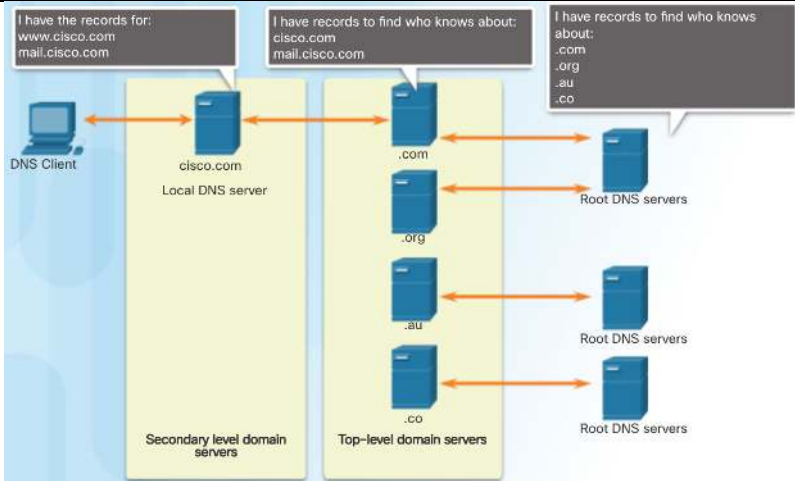
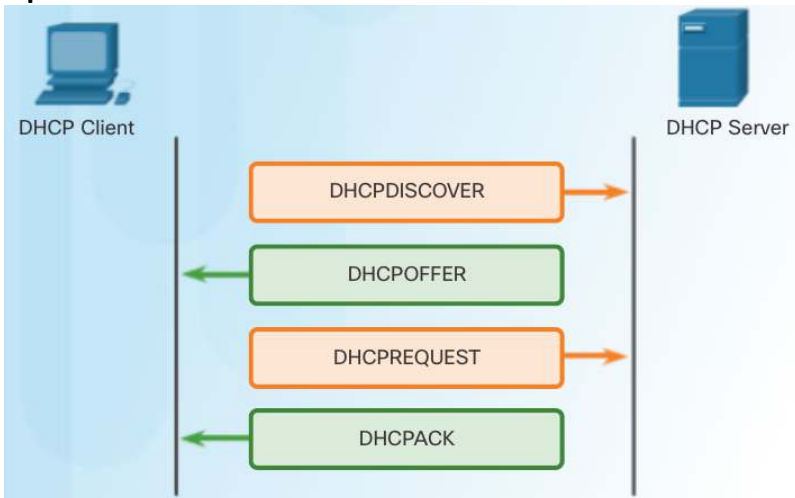
	Reliable delivery - Ensure all segments arrive even if it needs to be retransmitted Same-order delivery - Segments are reassembled into the proper order Flow control - Regulate amount of data the source transmits																																																																		
TCP Header	<table><tr><th colspan="4">Bit (0)</th><th>Bit (15)</th><th>Bit (16)</th><th colspan="4">Bit (31)</th></tr><tr><td colspan="4">Source Port (16)</td><td colspan="4">Destination Port (16)</td></tr><tr><td colspan="8">Sequence Number (32)</td></tr><tr><td colspan="8">Acknowledgement Number (32)</td></tr><tr><td>Header Length (4)</td><td colspan="2">Reserved (6)</td><td colspan="2">Control Bits (6)</td><td colspan="3">Window (16)</td></tr><tr><td colspan="4">Checksum (16)</td><td colspan="4">Urgent (16)</td></tr><tr><td colspan="8">Options (0 or 32 if any)</td></tr><tr><td colspan="8">Application Layer Data (Size varies)</td></tr></table>	Bit (0)				Bit (15)	Bit (16)	Bit (31)				Source Port (16)				Destination Port (16)				Sequence Number (32)								Acknowledgement Number (32)								Header Length (4)	Reserved (6)		Control Bits (6)		Window (16)			Checksum (16)				Urgent (16)				Options (0 or 32 if any)								Application Layer Data (Size varies)							
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UDP Features	- Lightweight transport protocol offering same segmentation and assembly as TCP but without reliability and flow control - Data reconstructed in order it is received - Lost segments are not resent - No session establishment - Does not inform send about resource availability - Stateless protocol (no tracking)																																																																		
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Having separate conversations	- Transport layer separates and manage multiple communications. - Both TCP and UDP manage multiple conversations by header fields to uniquely identify application through port numbers																																																																		
Port numbers	- Source port: originating application port - Destination port: Tell destination what service requested																																																																		
Socket pairs	- Source and destination port placed in segment - IP packet encapsulates segment - IP & Port number = socket																																																																		

LECTURE 9: Transport Layer	Types of port numbers	- IANA assigned 1. Well known ports (0 – 1023) 2. Registered Ports (1024 – 49151) 3. Dynamic or private ports
	TCP Connection establishment steps	1. The initiating client requests a client-to-server communication session with the server. 2. The server acknowledges the client-to-server communication session and requests a server-to-client communication session. 3. The initiating client acknowledges the server-to-client communication session.
	TCP Connection termination steps	1. When the client has no more data to send in the stream, it sends a segment with the FIN flag set. 2. The server sends an ACK to acknowledge the receipt of the FIN to terminate the session from client to server. 3. The server sends a FIN to the client to terminate the server-to-client session. 4. The client responds with an ACK to acknowledge the FIN from the server.
	TCP 3-way hand shake	TCP is full duplex protocol 1. Establishes that the destination device is present on the network 2. Verifies that the destination device has an active service and is accepting requests on the destination port number that the initiating client intends to use 3. Verifies that the destination device has an active service and is accepting requests on the destination port number that the initiating client intends to use 4. Informs the destination device that the source client intends to establish a communication session on that port number
	TCP reliability – ordered delivery	- Sequence number assigned in header of each packet - Represents first byte of data of TCP segment - ISN is set in session setup - As data transmitted, sequence number is incremented by no of bytes that have been transmitted - Missing segments can be identified
	Flow control	Window size/acknowledgments - 1460 bytes of data in each segment - Windows size agreed on during 3-way handshake Congestion avoidance - Can cause retransmission of lost TCP segments - Employs congestion handling mechanisms, timers
	UDP low overhead vs reliability	- Not connection oriented - No retransmission, sequencing and flow control - Functions not provide by transport are implemented elsewhere

	Applications that use TCP or UDP	TCP - HTTP - FTP - SMTP - Telnet	UDP - DHCP - DNS - SNMP - IPTV
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LECTURE 10: Application Layer

Application layer function	- Closest to end user - Exchange data between program on source and destination host
Presentation Layer function (3)	- Formatting at the source device into compatible form - Compressing data - Encrypting data
Session layer function	- Create and maintain dialogs between source and destination applications
TCP/IP model vs OSI model	
TCP/IP application layer protocols	Name system: DNS Host config: BOOTP, DHCP Email: SMTP, POP, IMAP File transfer: FTP, TFTP Web: HTTP, HTTPS
Client server model	- Client: request information, Server respond to request - All client and server processes are in application layer - Application protocols describe format of request and responses between client and server
Peer to peer networks (P2P)	- Data is access from peer device without use of dedicated server. (*Basics in lecture 1) - Involves two parts: - o Networks - o Applications
Common P2P applications	- E.g. Bitcoin, bit torrent, G2, eDonkey - Some are based on Gnutella protocol: Share whole files with other users - Many applications allow users to share pieces of file at the same time (Bit torrent)
HTTP & HTTPS	- Request/response protocol - 3 message types: GET – Client request for data. Request sent for HTML pages. POST – Upload files to web server PUT – Upload resources or content to webserver

LECTURE 10: Application Layer		HTTP not secure (in plain text) HTTPS is secure (authentication and encryption)
	Email protocols (3)	- Simple mail transfer protocol (send email) - Post office protocol (retrieve mail) - Internet message access protocol (retrieve mail)
	Domain name service (DNS)	- In data networks, Devices are labelled with numeric IP addresses to send and receive data over networks. - Domain names convert numeric address into simple recognisable name. - E.g. http://www.cisco.com vs 198.133.219.25
	DNS message format	DNS server stores different types of resource records A – End device IPv4 address NS – Authoritative name server AAAA – End device IPv6 address MX – Mail exchange record
	DNS hierarchy	
	DHCP	- Automation of IPv4 address, subnet mask, gateway and other parameter assignment - Addresses are leased for period of time and returned to pool for reuse - DHCP Employed for end user device. Static for networking Operation 

	FTP (File transfer protocol)	- Needs 2 connections between client & server: reply + command AND file transfer.
	Server message block (SMB)	- Client server file sharing protocol - Describes structure of shared network resources - It is a request response protocol (Printer) - Long term connection to servers